

What Do Wrist-Worn Stress Detectors Detect? An Untrained Arousal Index Matches Them Across Five Empatica E4 Datasets

Alvaro Octavio Ibarra Reyes and Chen Pan

Abstract—Wrist-worn stress detectors are trained to separate stressor periods from rest and are reported as detecting stress. We ask what they detect. On five public laboratory Empatica E4 datasets (132 subjects) with audited protocol-stage labels, subject-grouped leave-one-dataset-out evaluation and corrected tests, a heart-rate, heart-rate-variability and electrodermal model trained on four datasets loses 0.015–0.05 balanced accuracy on the fifth (none significant; upper 95% bounds 0.08–0.13). An untrained index, the mean of per-subject z-scored heart rate and tonic electrodermal activity with a threshold set on the other datasets, performs as well: no model-minus-index difference survives Holm correction. At matched arousal the model separates stress from baseline and rest only weakly (AUROC 0.58), and no better from a lower-demand social task, hyperventilation, Lego assembly or anger clips; these nulls exclude only deviations above 0.12. Exercise, the one unambiguous non-stress state, is called stress in 69% of windows until it is added to training (8–10%). Minutes since recording start alone predicts stress better than physiology on the two hardest datasets, whose baselines coincide with sensor settling. Domain adaptation, few-shot personalisation and per-user thresholds add nothing significant. On current public E4 data, wrist stress detectors behave as arousal detectors. We propose an index-only baseline, a specificity panel and a matched-arousal test as reporting standards, and release audited labels, frozen subject splits and code.

Index Terms—Wearable stress detection, Empatica E4, autonomic arousal, cross-dataset generalization, specificity, electrodermal activity, heart rate variability.

I. INTRODUCTION

STRESS detection from wrist wearables is usually posed as a classification problem: windows recorded during a laboratory stressor are labelled stress, windows recorded during rest or a control task are labelled non-stress, and a model is trained to separate them. Reported accuracies on the standard datasets are high, and the models are described as detecting stress. Two observations motivate this paper. First, the cross-dataset literature disagrees about whether such models transfer: Prajod et al. [1] and Calza-Metre and Borzi [2] report large drops when the stressor changes, Xiao et al. [3] report balanced accuracy falling from 0.88 to 0.53–0.69 across seven wrist datasets, whereas Mishra et al. [4] found a cost of about 0.01 between two Empatica E4 studies with the same stressor, and Kwon et al. [5] found near-perfect transfer to WESAD but near-chance results on Stress-Predict. Second, the wrist device measures pulse rate, a weak vagal proxy, wrist electrodermal

activity (EDA), skin temperature and movement. None of these is stress; all of them respond to autonomic arousal from any cause. Kwon et al. [5] matched heart rate within subject and found that WESAD stress remains separable from baseline through EDA and temperature, which rules out a heart-rate shortcut there. Whether a detector separates stress from other arousing non-stress states when heart rate and EDA are matched together has not, to our knowledge, been tested.

We address both observations with the same five public E4 datasets. After auditing every dataset’s labels against its descriptor paper, validating wrist cardiac features against chest electrocardiography (ECG), and fixing an evaluation protocol with subject-grouped splits and corrected tests, we show that the transfer cost is small once labels and per-subject normalisation are right, and that the remaining disagreement in the literature is largely about absolute difficulty rather than transfer. We then ask what transfers. An untrained arousal index is scored under the same leave-one-dataset-out protocol as the model; a specificity panel scores the fixed detector on states absent from training; and a matched-arousal test asks whether stress stays separable from those states, and from rest, at equal arousal. A time-only classifier, a sensor-settling analysis and a test-side error budget then locate the remaining within-dataset ceiling.

The contributions are: (1) a five-dataset leave-one-dataset-out benchmark with audited protocol-stage labels, showing a small, non-significant transfer cost (0.015–0.05 balanced accuracy, upper bounds 0.08–0.13) and diagnosing one apparent domain gap as a missing channel; (2) evidence that the trained detector behaves as an arousal detector: an untrained heart-rate and EDA index transfers as well, stress is only weakly separable from rest at matched arousal and no better from lower-demand tasks, and exercise is called stress until it is in training; (3) a quantification of time in session and sensor settling as confounds, with a position-matched evaluation rule; (4) three reporting checks, an index-only baseline, a specificity panel and a matched-arousal test with a reference pair, together with an error budget and leakage-controlled negative results for domain adaptation, few-shot personalisation and per-user thresholds. Labels, frozen splits and scripts are released. Section II reviews related work, Section III describes the datasets and the label audit, Section IV the features, normalisation and evaluation rules, Section V the results, and Section VI discusses implications, reporting norms and limitations.

The authors are with the Department of Electrical and Computer Engineering, The University of Texas at San Antonio, San Antonio, TX 78249 USA (e-mail: [email]@utsa.edu; [email]@utsa.edu).

II. RELATED WORK

A. Cross-dataset transfer

Kwon et al. [5] ran pooled leave-one-dataset-out (LODO) evaluation on three E4 corpora with logistic regression and gradient boosting and no per-subject normalisation, obtaining an area under the receiver operating characteristic curve (AUROC) of about 0.90 with WESAD as target but 0.52–0.57 on Stress-Predict and the Nurse dataset; their within-dataset AUROC on Stress-Predict was already 0.63, so the transfer cost there is 0.06–0.08. Calza-Metre and Borzi [2] report pairwise zero-shot transfer across four E4 corpora, with WESAD macro-F1 falling from 0.87 within to 0.63 from Campanella. Xiao et al. [3] train on one or two of seven wrist datasets and observe balanced accuracy 0.53–0.69 out of distribution. Mishra et al. [4], with per-participant normalisation and matched arithmetic stressors, found a median AUROC cost of about 0.01. Schreiber and Maleshkova [6] pool three datasets under leave-one-subject-out (LOSO) evaluation and name LODO as future work. Prajod et al. [1] attribute transfer failure to stressor type, on chest ECG. Our reading of this literature is that the reported failures come from low within-dataset ceilings, from stressor mismatch, or from the absence of per-subject normalisation, and the present study tests each.

B. Label quality and confounds

Several authors name label noise as a cause of poor transfer [5], [7] but none audits and fixes labels and measures the effect. Mishra et al. [4] exclude post-stressor rest because it carries residual arousal; Aydoğan and Villagra Povina [8] found independently that the PhysioNet stress sessions mix rest and stressor blocks. Richer et al. [9] note that comparing a stressor to a pre-stress baseline confounds the comparison with sequence effects. Tognotti et al. [10] show that test-inclusive baseline normalisation inflates balanced accuracy by 3–13 points, and Fecke and Rehof [11] normalise from a calibration baseline on chest ECG. Aydoğan and Villagra Povina [8] report that an XGBoost model labels 82.6% of PhysioNet exercise windows as stress, and 6–7% once exercise is a training class. Saeb et al. [12] and Bahameish et al. [15] document the inflation from record-wise cross-validation. Schmidt et al. [13] recommend leave-one-subject-out validation and treat physical activity as a confound for which it is “sufficient to estimate the intensity level of an activity”. Vos et al. [14], found that apart from two studies, including Mishra et al. [4], none of the reviewed work tested its models on a totally unseen dataset, and estimate that “at least 34 test subjects would be required to achieve 80% statistical power”.

C. Personalisation and the arousal question

Adding a few labelled windows of a new user to a generic model is well established [16], [17], [18], [19]. Where calibration windows are taken chronologically, gains are small: Tervonen et al. [18] found retraining XGBoost gained 1.7 balanced-accuracy points on WESAD with 1–3 windows per class, and Akkaya [19] reports that the few-shot benefit did not survive a prospective chronological protocol. On wrist

cardiac validity, Milstein and Gordon [20] found that the root mean square of successive differences (RMSSD) from the E4 correlated with ECG at $r = 0.42$ during a dyadic conversation against 0.74–0.81 at rest, and Watanabe et al. [21] found the Trier Social Stress Test (TSST) to be the worst condition for E4 beat detection on WESAD. Sosa et al. [22], reviewing multimodal arousal classifiers, state that “physiological signals primarily index arousal and regulatory processes rather than affective valence” and that sympathetic activation “can therefore arise not only during stress, but also during non-stressful states”; the reviewed studies did not test classifiers on such states. Kwon et al. [5] balanced heart rate within 5-bpm bins per subject: on WESAD the all-feature AUROC moved from 0.955 to 0.938 and EDA plus temperature alone kept 0.935, while on Stress-Predict every matched feature set stayed weak (EDA plus temperature 0.607). Their contrast is stress against baseline; the matched-arousal test below instead matches heart rate and EDA jointly and contrasts stress with other arousing states.

The gap is therefore twofold. No study has scored one fixed, subject-independent wrist detector across a panel of non-stress arousal states, or asked whether stress remains separable from those states at equal arousal; and no study has decomposed the within-dataset ceiling of wrist stress detection into recovery, sensor, onset, responder and stressor terms. The present paper addresses both on five public datasets with audited labels.

III. DATASETS AND LABEL AUDIT

Five public datasets recorded with the Empatica E4 (blood volume pulse (BVP) 64 Hz, EDA 4 Hz, skin temperature 4 Hz, accelerometer 32 Hz) contain both stress and non-stress periods: WESAD [23] (15 subjects; TSST speech and arithmetic before a panel, standing), the PhysioNet stress and exercise dataset of Hongn et al. [24] (35; Stroop and the Trier Mental Challenge Test (TMCT) of mental arithmetic under time pressure, seated, plus aerobic and anaerobic sessions; its opinion tasks yield no complete window), Stress-Predict [25] (34 after exclusion; Stroop and a TSST-style interview, plus a hyperventilation block that its descriptor lists among the stress-inducing tasks), UBFC-Phys [26] (19; speech and arithmetic in a harder “test” version, a job interview and a countdown in steps of 17, or an easier “control” version, a holiday speech and a countdown in steps of 10, both spoken aloud to the experimenter; the authors note that both versions may “imply a high stress level”) and Campanella 2024 [27] (29; backward subtraction, plus three Lego assembly tasks designed to create work-like “mental strain”, one with a spoken countdown; the closing CV presentation lies in the untagged tail and is excluded). EPM-E4 [28] (33 subjects with clip timing) supplies film-induced anger, fear, happiness and sadness and is used only as a probe. Table I gives the harmonised classes.

Every labelling decision was checked against the dataset’s descriptor paper. Seven corrections followed: in WESAD, labels had to be taken from the synchronised file rather than mapped by length onto the raw E4 export, whose recording starts 16–26 min earlier; in PhysioNet, only the tagged stressor tasks count as stress, not the whole stress session, one subject

TABLE I
HARMONISED CLASSES (SUBJECTS / 60-S WINDOWS AT 30-S STEP)

Dataset	Original label → harmonised class	Subj./win.
WESAD	TSST → Stress	15 / 311
	Baseline; Meditation; Amusement	15 / 566; 356; 165
PhysioNet	Stroop, TMCT, subtraction → Stress	35 / 232
	Baseline (3 min before first tag); First and second rest	35 / 174; 1193
	Aerobic; Anaerobic (whole sessions)	30 / 2143; 31 / 1620
	Stroop; TSST interview → Stress	33 / 851
Stress-Predict	Baseline; Relax → Rest	34 / 616; 1079
	Hyperventilation (probe only)	31 / 53
	T2, T3 test group → Stress	11 / 110
	T1 rest → Baseline	19 / 95
UBFC-Phys	T2, T3 control group → Control task (probe)	8 / 80
	Backward subtraction → Stress	29 / 102
Campanella	Baseline	29 / 116
	Lego tasks → Manual task (probe)	29 / 667
EPM-E4	Anger; Fear; Happiness; Sadness (probes)	33 / 33; 394; 297; 131

with a documented bad fit (f13) is excluded, and the 3 min before the first tag serve as baseline; in Stress-Predict, subject S01 violated the protocol and is dropped, and task boundaries whose tags are more than 180 s from the logged time are dropped; in Campanella, which logs no per-subject task markers, the subtraction span ends at a per-subject end estimated from the recording length. In addition, the Stress-Predict descriptor tags its 2-min hyperventilation block as a “stress-inducing task”; it is a respiratory manoeuvre with no evaluative component, and we keep it as a separate probe class. The three participants recorded in two files are treated as one subject each, so that no person appears on both sides of a split.

A cautionary note on evaluation. An earlier version of this pipeline used window-level train/test splitting with 50%-overlapping windows and reported 94% accuracy on a six-class task (92% in our reproduction); the same model scores 49% (balanced accuracy 0.36) when 15% of subjects are held out (`leakage_check.csv`). Record-wise splitting [12] remains common in this field, and every number below uses subject-grouped splits.

IV. METHODS

A. Features

Windows are 60 s at a 30-s step, anchored inside one protocol stage. Cardiac features, the mean and standard deviation (SD) of heart rate (HR) and three heart-rate-variability (HRV) measures (RMSSD, the SD of normal-to-normal intervals (SDNN) and the proportion of successive differences above 50 ms (pNN50)), come from BVP beats detected with NeuroKit2 after band-pass filtering and artefact rejection; windows with fewer than 50% usable beats carry missing cardiac values. Against WESAD chest ECG on the same 1,398 windows, wrist heart rate has a mean absolute error of 2.4 bpm ($r = 0.94$);

RMSSD is weak ($r = 0.53$, bias +34 ms), and only 37% of TSST windows keep usable photoplethysmography (PPG) because speech moves the wrist (heart-rate bias -7 bpm there). EDA features are mean, SD, range, slope, tonic and phasic means, and skin conductance response (SCR) count and amplitude. Temperature features are included only in the “physiology” set; the main set is HR/HRV/EDA, for reasons given in Sections V-B and V-E. Accelerometer features are used only in the modality comparison.

B. Normalisation

Features are z-scored per subject using all windows of that subject’s stress-protocol recording, without labels. This is transductive; two non-transductive variants are reported as sensitivity analyses: raw features, and causal scaling from strictly earlier windows only (the first four windows share the statistics of that opening period).

C. Task, models and evaluation

The main task is stress versus all non-stress states (baseline, rest, meditation, amusement). Baseline-versus-stress alone is confounded with time in session (Section V-E). The classifier is XGBoost with fixed hyperparameters; a nested-tuning comparison of XGBoost, random forest and a multilayer perceptron found one significant difference in 72 comparisons after Holm correction, on the nine-class affective task only and none on the stress tasks (`significance_models.csv`), so tuning is not repeated. Evaluation uses 20 subject-grouped splits with 20% of each dataset’s subjects held out, LOSO where per-subject scores are needed, and LODO in which the target’s held-out subjects are scored by a model trained (a) on the target’s other subjects, (b) on the other four datasets only, or (c) on both. Paired comparisons use Nadeau–Bengio corrected t -tests [29] with Holm correction; per-subject rates carry subject-bootstrap 95% confidence intervals (CI). The threshold is 0.5 throughout; balanced accuracy (BA) and AUROC are reported.

D. Four evaluation rules

Specificity panel. A fixed detector is scored on non-stress states absent from its task definition, reporting the false-stress rate ($p > 0.5$) under three training conditions: the state’s dataset absent from training; all datasets in training but the state itself absent; the state added to the training negatives. *Matched-arousal test.* A label-free arousal index (mean of per-subject z-scored heart rate and tonic EDA, or tonic EDA alone where heart rate is missing) is computed per window; stress windows are pooled with one non-stress arousal state, binned into arousal quintiles, and the AUROC of the stress score is computed within bins and averaged by bin size. An AUROC near 0.5 means the model has learned nothing beyond arousal magnitude for that pair. *Position-matched negatives.* Non-stress test windows are restricted to the 5th–95th percentile range of minutes-since-start spanned by that dataset’s stress windows. *Test-side error budget.* Filters applied to out-of-sample predictions without retraining, with gains Shapley-averaged over all filter orders.

TABLE II
LEAVE-ONE-DATASET-OUT BALANCED ACCURACY ON HELD-OUT
SUBJECTS, STRESS VS ALL NON-STRESS (MEAN OF 20 SPLITS)

Target	HR/HRV/EDA model			Index		+ temperature		
	Within	Other 4	All 5	Other 4	Within	Other 4	All 5	
WESAD	0.916	0.884	0.921	0.836	0.900	0.840	0.876	
PhysioNet	0.776	0.746	0.783	0.743	0.760	0.735	0.770	
Stress-Predict	0.697	0.646	0.689	0.657	0.712	0.645	0.699	
UBFC-Phys	0.877	0.837	0.839	0.762	0.877	0.589	0.867	
Campanella	0.882	0.867	0.895	0.877	0.897	0.827	0.895	

Index: the untrained arousal index (mean of per-subject z-scored heart rate and tonic EDA), thresholded at the value that maximises BA on the other four datasets.

V. RESULTS

A. Subject-independent performance

On the pooled baseline-versus-stress task of WESAD and PhysioNet (50 subjects) with raw (unnormalised) features and the fixed-hyperparameter model, 20 subject-grouped splits give BA 0.83 (Nadeau–Bengio 95% CI 0.79–0.88) with all modalities, 0.81 (0.76–0.86) with physiology only and 0.75 (0.68–0.82) with the accelerometer only; LOSO agrees (0.85, 0.81, 0.75). No modality difference survives Holm correction ($p_{\text{holm}} = 1.0$, `significance_modalities.csv`), and a complete 108-configuration nested tuning run changes none of these conclusions. The model is therefore not detecting movement alone, and physiology carries most of the signal. On the main task and all five datasets, within-dataset BA with HR/HRV/EDA features is 0.916 (WESAD), 0.882 (Campanella), 0.877 (UBFC-Phys), 0.776 (PhysioNet) and 0.697 (Stress-Predict) (Table II). The two lowest ceilings belong to the two datasets whose stressors provoke the smallest electrodermal response: mean EDA rises by 0.4 z in PhysioNet and 0.7 z in Stress-Predict against 1.7–2.2 z in the other three (`stress_effect_size_all_datasets.csv`).

B. Cross-dataset transfer

Table II and Fig. 1 give the LODO result. With HR/HRV/EDA features, a model trained on the other four datasets loses 0.03 (WESAD), 0.03 (PhysioNet), 0.05 (Stress-Predict), 0.04 (UBFC-Phys) and 0.015 (Campanella) BA against the within-dataset model on the same held-out subjects. Training on all five datasets matches or exceeds within-dataset training on WESAD, PhysioNet and Campanella and is within 0.04 on the other two (Stress-Predict 0.689 against 0.697, UBFC-Phys 0.839 against 0.877). None of the five costs is significant after Holm correction (Nadeau–Bengio $p_{\text{holm}} \geq 0.20$; only Stress-Predict excludes zero before correction, 0.052 [0.003, 0.101]), but with 3–7 held-out subjects per split the upper 95% bounds reach 0.08–0.13 (`D_lodo_cost_corrected.csv`). We therefore claim no large cost, not equality.

An untrained arousal index transfers as well as the model. If the detector has learned arousal magnitude, a label-free arousal index should do nearly as well. The index (mean of per-subject z-scored heart rate and tonic EDA, EDA alone where heart rate is missing) needs no training; only its threshold is set, on the other four datasets. Table II gives its external

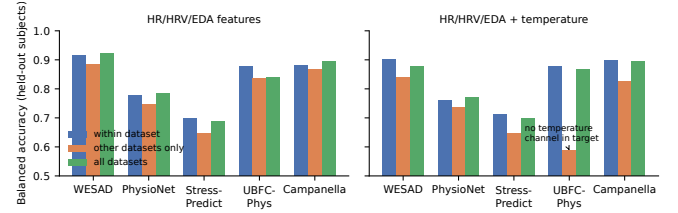


Fig. 1. Leave-one-dataset-out balanced accuracy on held-out subjects for the two feature sets. With temperature included, UBFC-Phys, which records no temperature, drops to 0.589 when trained on the other datasets although its AUROC stays at 0.931.

BA: 0.836 on WESAD, 0.743 on PhysioNet, 0.657 on Stress-Predict, 0.762 on UBFC-Phys and 0.877 on Campanella, with AUROC 0.73–0.95. The trained model beats it by 0.048 on WESAD and 0.075 on UBFC-Phys, ties on PhysioNet (+0.003), and trails it on Stress-Predict and Campanella; no difference survives Holm correction (paired Nadeau–Bengio, $p_{\text{holm}} \geq 0.14$; `C_arousal_index_lodo.csv`). An EDA-only index loses significantly to the model on PhysioNet (−0.15) and a heart-rate-only index on UBFC-Phys and Campanella, so neither channel alone suffices. Up to the power of these tests, a model trained on four datasets of stressor labels behaves on the fifth like an untrained two-channel arousal index.

A missing channel posing as a domain gap. With temperature included, external UBFC-Phys falls to 0.589 BA while its AUROC stays 0.931 and an oracle threshold recovers 0.910 (`threshold_transfer_probe_summary.csv`). UBFC-Phys has no temperature in any window, so XGBoost’s missing-value branch pushes its mean stress score to 0.18 against 0.31–0.42 on the other targets. Ranking is intact; the threshold is not. Dropping temperature lifts external UBFC-Phys to 0.837. Elsewhere the remaining external loss is mostly ranking: an oracle threshold recovers only 0.02–0.05 with HR/HRV/EDA features, and a label-free cut matching the source stress rate hurts Campanella (0.867 to 0.732).

Normalisation. The small cost does not depend on transductive scaling. With raw features the external cost is −0.01 to 0.05 BA; causal scaling keeps it on WESAD (0.845 within, 0.841 external), PhysioNet (0.774, 0.744) and Stress-Predict (0.688, 0.665) but loses about 0.10 externally on UBFC-Phys and Campanella, whose recordings are short (7–11 windows per subject). Whole-session scaling mainly raises the absolute level on both sides. No external difference between variants survives Holm correction across 60 tests.

Label fixes. Every external score rose after the seven corrections (for example UBFC-Phys 0.790 to 0.837, whose own data did not change), but on the two datasets where a paired test is possible only 1 of 16 differences survives Holm correction (WESAD external AUROC, +0.0075, $p_{\text{holm}} = 0.021$). In the two-dataset setting of Kwon et al. [5] (WESAD and Stress-Predict), the fixes change transfer by at most 0.02; our pipeline nevertheless reaches external AUROC 0.67 on Stress-Predict against their 0.56, and per-subject scaling raises WESAD as target from 0.834 to 0.914 AUROC. The label audit is a correctness contribution; the small transfer cost comes from

TABLE III
FALSE-STRESS RATE ON STATES ABSENT FROM THE TASK DEFINITION
(HR/HRV/EDA; SUBJECT-BOOTSTRAP 95% CI)

Probed state (subjects / windows)	Dataset absent	State absent	Added as negative
PhysioNet aerobic (30 / 2143)	0.69 [0.59, 0.80]	0.61 [0.52, 0.71]	0.10 [0.05, 0.15]
PhysioNet anaerobic (31 / 1620)	0.70 [0.61, 0.79]	0.58 [0.50, 0.66]	0.08 [0.05, 0.11]
UBFC-Phys control version (8 / 80)	0.55 [0.46, 0.66]	0.57 [0.48, 0.66]	0.56 [0.45, 0.67]
Stress-Predict hyperventilation (31 / 53)	0.42 [0.26, 0.59]	0.54 [0.36, 0.71]	0.55 [0.36, 0.72]
Campanella Lego tasks (29 / 667)	0.43 [0.38, 0.48]	0.42 [0.36, 0.48]	0.33 [0.27, 0.39]
EPM-E4 anger (33 / 33)	0.42 [0.27, 0.61]	0.44 [0.27, 0.61]	0.35 [0.18, 0.53]
EPM-E4 fear (33 / 394)	0.35 [0.25, 0.45]	0.33 [0.23, 0.43]	0.25 [0.17, 0.34]
EPM-E4 sadness (33 / 131)	0.37 [0.27, 0.49]	0.35 [0.24, 0.49]	0.30 [0.19, 0.42]
EPM-E4 happiness (33 / 297)	0.24 [0.14, 0.33]	0.25 [0.15, 0.38]	0.18 [0.09, 0.29]
Reference: in-task non-stress states	0.08–0.29	0.05–0.25	—
Reference: stress recall, by dataset	0.53–0.91	0.63–0.95	within –0.05

features, task definition and normalisation.

C. Specificity panel

The LODO task keeps only baseline, rest, meditation and amusement as negatives; other recorded states were never scored. Table III and Fig. 2(a) give their false-stress rates. Exercise is the one probed state that is unambiguously not stress. A model that never saw exercise calls 69–70% of PhysioNet exercise windows stress, matching the 71–83% of Aydoğan and Villagra Povina [8]; adding exercise to the training negatives collapses this to 8–10%, and the external PhysioNet drop is significant (BA 0.746 to 0.583, $p_{\text{holm}} = 0.011$). No other state behaves this way, but the other probes are not clean non-stress states. The UBFC-Phys control version is an easier social task that its authors expect to be stressful, the Campanella Lego tasks were built to induce work-like strain, the Stress-Predict descriptor lists hyperventilation among its stress-inducing tasks, and the EPM-E4 clips are passive viewing. The UBFC-Phys control version is called stress at 0.55–0.57 whether or not the model has seen it; hyperventilation rises from 0.42 to 0.54–0.55 once Stress-Predict is in training; Lego tasks fall only from 0.43 to 0.33; and the EPM-E4 clips fall by 0.05–0.10. Adding a probed state to the negatives changes stress recall by at most –0.05 (UBFC-Phys 0.74 to 0.70 with Lego added; PhysioNet 0.74 to 0.71 and Campanella 0.92 to 0.89 with the EPM-E4 clips added), and exercise negatives raise it (Stress-Predict 0.63 to 0.70, UBFC-Phys 0.74 to 0.79), so the failure is not a sensitivity trade-off. The hyperventilation (1.7 windows per subject), anger (one window per subject) and UBFC-Phys control (8 subjects) cells are small; Lego and the fear clips carry the statistics.

D. Matched-arousal test

Table IV and Fig. 2(b) give the result. The reference row sets the scale: at matched arousal, even the trained contrast, stress against baseline and rest, reaches only 0.584 for the external model. The wrist features therefore carry little stress

TABLE IV
MATCHED-AROUSAL TEST: AUROC OF THE STRESS SCORE INSIDE AROUSAL QUINTILES, AND ITS DIFFERENCE FROM THE BASELINE/REST REFERENCE UNDER THE SAME MODEL

Stress versus	n	Within-bin [CI]	Δ ref [CI]	p_{holm}	MDD
<i>External model (probed dataset absent from training)</i>					
Baseline and rest (ref.)	3839	0.584 [0.54, 0.64]	—	0.004	0.07
Exercise	3763	0.759 [0.67, 0.82]	+0.18 [+0.08, +0.25]	0.004	0.12
Fear clips	394	0.633 [0.58, 0.69]	+0.05 [0.00, +0.10]	0.004	0.07
UBFC-Phys control	80	0.577 [0.50, 0.67]	–0.01 [–0.10, +0.09]	0.41	0.12
Hyperventilation	53	0.523 [0.44, 0.61]	–0.06 [–0.15, +0.02]	0.62	0.12
Anger clips	33	0.390 [0.31, 0.52]	–0.19 [–0.27, –0.06]	0.41	0.15
Lego tasks	667	0.355 [0.24, 0.54]	–0.23 [–0.34, –0.03]	0.13	0.20
<i>Model trained with the probed state as a negative (ref. 0.71–0.72)</i>					
Exercise	3763	0.892 [0.84, 0.93]	+0.18 [+0.13, +0.23]	0.003	0.07
Fear clips	394	0.779 [0.74, 0.82]	+0.05 [+0.01, +0.10]	0.003	0.06
UBFC-Phys control	80	0.568 [0.46, 0.67]	–0.16 [–0.26, –0.05]	0.64	0.15
Hyperventilation	53	0.560 [0.45, 0.67]	–0.16 [–0.27, –0.06]	0.64	0.15
Lego tasks	667	0.496 [0.44, 0.70]	–0.22 [–0.28, –0.01]	0.94	0.22
Anger clips	33	0.431 [0.33, 0.62]	–0.29 [–0.38, –0.12]	0.64	0.20

Within-bin AUROC is weighted by bin size, with a subject-bootstrap CI (2000 draws). Δ ref: probe-pair AUROC minus the baseline/rest pair under the same model, joint subject bootstrap within dataset. p_{holm} : permutation test of the within-bin AUROC against 0.5 (labels shuffled across subject-by-class cells within each bin), Holm-corrected within each model. MDD: deviation from 0.5 detectable with 80% power (2.8 bootstrap SD). Only hyperventilation (external) is equivalent to 0.5 within ± 0.10 (90% CI [0.45, 0.60]); no pair is within ± 0.05 (reviewer_reanalyses/).

information beyond arousal magnitude, whatever the comparator. Against that reference, the UBFC-Phys control version (0.577, $\Delta = -0.01$) and hyperventilation (0.523, $\Delta = -0.06$) are indistinguishable from the reference, and anger clips and Lego tasks fall below it. Once the model has been trained with the probed state as a negative, the reference rises to 0.71–0.72, but the UBFC-Phys control version, hyperventilation, Lego and anger stay at 0.43–0.57, 0.16–0.29 below it with CIs excluding zero: the model learns to separate stress from rest beyond arousal, but not from these demanding tasks. These are absence-of-evidence results. Only hyperventilation is equivalent to chance within ± 0.10 , and the tests could detect only deviations of 0.12–0.22 from 0.5 at 80% power. Two states are separable. Exercise reaches 0.892 within bins once the model has seen it ($p_{\text{holm}} = 0.003$) and 0.759 even without, and fear clips reach 0.779; the bootstrap difference between the exercise AUROC and each non-separable pair excludes zero (lower bounds 0.18–0.36). For that model, repeating the test with a heart-rate-only index gives 0.91 and with an EDA-only index 0.82, so exercise stays separable whichever channel defines arousal; it is separated by its pattern across channels, not by arousal level. (For the external model, which has not seen exercise, the same indices give 0.65 and 0.54.)

Because the index is built from two model features, the test was repeated with index and model on disjoint channels (partA_disjoint.csv): a heart-rate-only index with an EDA-only model, and an EDA-only index with a cardiac-only model. Hyperventilation and the UBFC-Phys control version stay inseparable in every cell (0.45–0.54). The other pairs depend on the variant. Lego becomes separable under the heart-rate index (0.73 external, 0.84 once trained), where the pair shrinks to the 259 stress and Lego windows with usable pulse (63% of Campanella windows have none); anger becomes separable under the EDA index once trained (0.74);

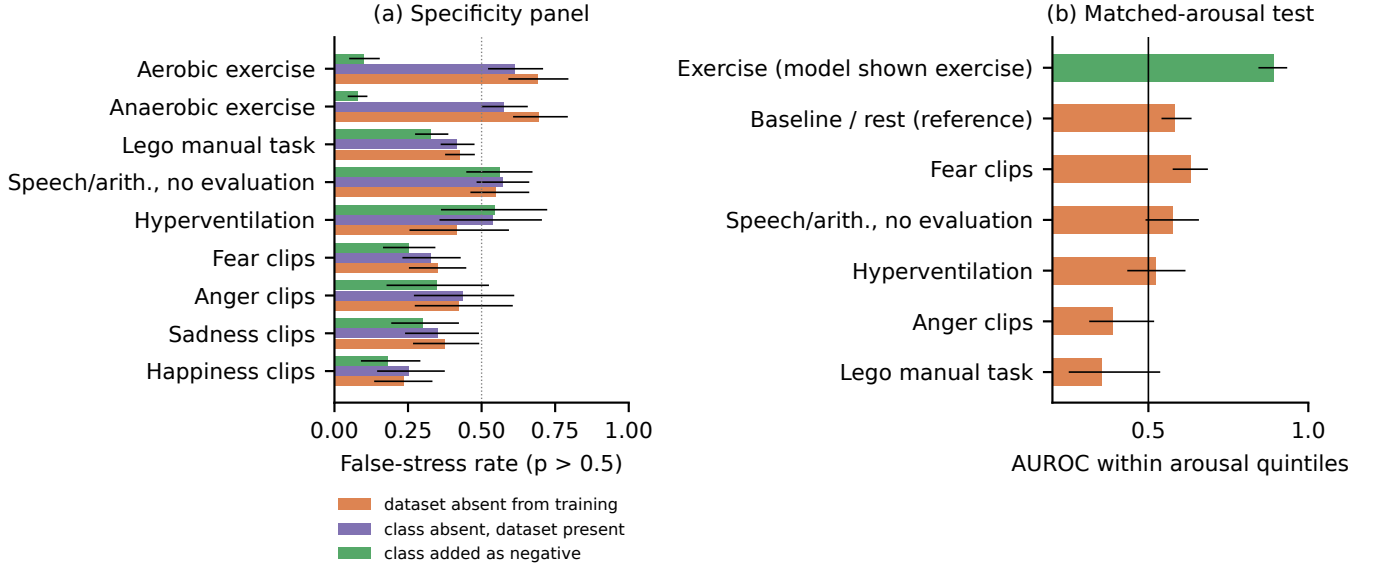


Fig. 2. (a) Specificity panel: false-stress rate of the HR/HRV/EDA detector on states absent from its task definition, under three training conditions, with subject-bootstrap 95% CIs. (b) Matched-arousal test: AUROC of the stress score within arousal quintiles for stress versus each state (external model; for exercise, the model that has seen exercise). Only exercise and fear clips remain separable at equal arousal; the baseline/rest reference itself reaches only 0.58 externally.

and with the EDA index the external cardiac-only model scores exercise above stress (AUROC 0.29), since within EDA bins exercise windows carry far higher heart rate (median heart-rate index 3.9 against 1.2). Removing only the three features closest to the index (`hr_mean`, `eda_tonic_mean`, `eda_mean`) leaves correlated EDA features in the model and changes no row by more than 0.09. Heart rate is also missing unevenly: in WESAD 63% of stress windows have no usable pulse against 10% of baseline windows, so the index falls back to EDA for most WESAD stress windows. Exercise windows sit almost entirely above every stress window (median index 4.8 against 0.8), so its matched bins cover only the lowest three quintiles.

E. Time in session and sensor warm-up

Baseline is recorded first in every dataset, so any baseline-referenced comparison is confounded with time in session. Fig. 3(a) quantifies this: minutes since recording start, used as the only feature, reaches BA 0.83 on PhysioNet (physiology 0.78; better in 16 of 20 splits) and 0.95 on Stress-Predict (physiology 0.70; 20 of 20), ties on WESAD (0.90 against 0.92) and is perfect on Campanella. Given minutes as an extra feature, the within-dataset model exploits it (Stress-Predict 0.70 to 0.95) and then collapses externally (0.66 to 0.56; WESAD 0.90 to 0.45), because protocol timing does not transfer. Any within-dataset pipeline that leaks time, through raw temperature, drift features or the absence of per-subject normalisation, inherits this. The physiology model survives position-matched negatives with changes of at most -0.03 within and -0.04 externally (`time_probe_summary.csv`), so the LODO result is not riding on time; the matched negatives are mostly

post-stressor rest, so this also bounds recovery contamination. A baseline-referenced feature set, by contrast, calls 50–83% of later non-stress windows stress across the WESAD–PhysioNet pair (`order_confound_check.csv`).

Time is informative on these datasets because baseline coincides with the warm-up phase of the sensor. Measured from the start of each E4 recording (`warmup_since_donning.csv`), subject-centred skin temperature rises at 0.02 – 0.08 °C/min in the first 10 min on all four datasets with a continuous clock keeps rising past 20 min on WESAD and rises until about minute 20 on Campanella before levelling off ($+1.8$ and $+0.9$ °C between minutes 1–5 and 30–40; 13 of 15 WESAD subjects warm over the first 30 min, by up to $+6.5$ °C), and tonic EDA rises in the first 10 min on all four. Baseline falls inside that phase on Campanella (100% of baseline windows within 10 min of recording start), Stress-Predict (97%) and PhysioNet (68% within 10 min, 90% within 15), and stress arrives at a median of 7–28 min. WESAD is the exception: its synchronised protocol starts 16–26 min after the raw E4 recording (median 21 min, found by matching the wrist EDA stream), so its baseline begins 17–49 min after the device was switched on (median 33 min) and no baseline window falls within 15 min of recording start. WESAD is also the one dataset on which the time-only classifier does not beat physiology. Whether recording start equals donning is not documented by any dataset; on WESAD, every subject’s first minute of temperature reads 4 – 6 °C above skin, which shows the device was placed within that minute. This is the quantitative form of the sequence effect described by Richer et al. [9] and the reason temperature features track time and are excluded from the main feature set.

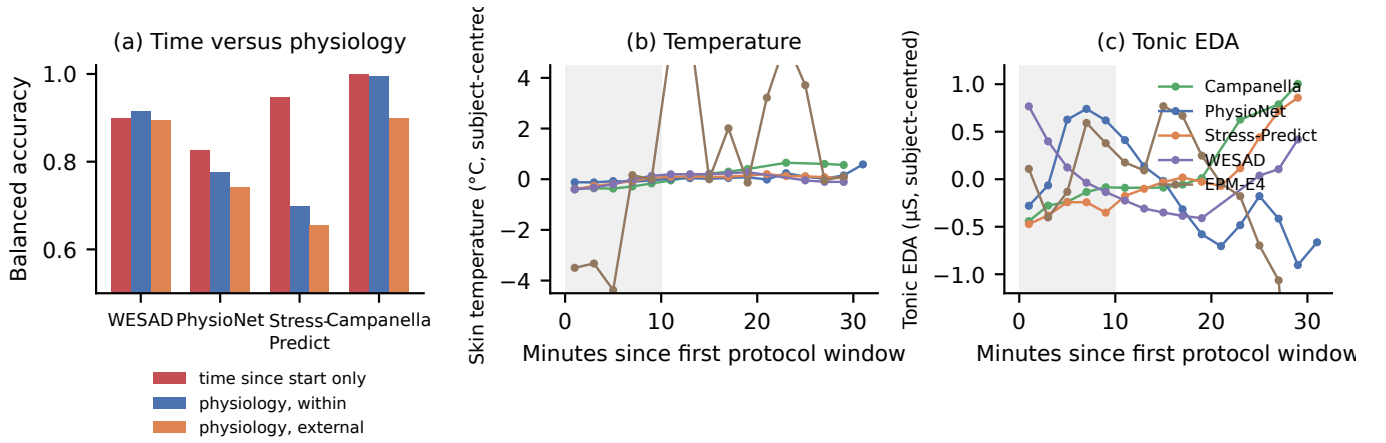


Fig. 3. (a) Balanced accuracy of a classifier that sees only minutes since recording start, against the physiology model within and across datasets. (b), (c) Subject-centred skin temperature and tonic EDA against minutes since the first protocol window (2-min bins); the shaded band marks the first 10 min, in which 100% of Campanella, 97% of Stress-Predict and 68% of PhysioNet baseline windows fall when measured from recording start (90% within 15 min). WESAD’s protocol clock starts 16–26 min after its recording; EPM-E4’s starts at recording.

TABLE V
TEST-SIDE ERROR BUDGET: BALANCED-ACCURACY GAIN OF EACH FILTER, SHAPLEY-AVERAGED OVER 24 ORDERS (HR/HRV/EDA)

Dataset, model	All	F1	F2	F3	F4	Resid.
PhysioNet, within	0.776	0.003	0.007	0.020	0.020	0.175
PhysioNet, external	0.746	0.015	0.006	0.036	0.008	0.190
Stress-Predict, within	0.710	0.019	0.011	0.018	0.012	0.229
Stress-Predict, external	0.653	0.015	−0.015	0.012	0.005	0.329
WESAD, external	0.887	0	0.045	0	0.002	0.066

F. Error budget and the chest-ECG bound

Table V and Fig. 4 test four explanations for the low ceilings on PhysioNet and Stress-Predict as test-side filters on LOSO and external predictions: F1 drops rest windows within 5 min of a stressor offset (recovery contamination); F2 keeps windows with cardiac coverage of at least 0.5 (sensor loss); F3 drops the first 60 s of each stressor (onset latency); F4 keeps only subjects whose own stressor-minus-baseline heart rate or tonic EDA exceeds 0.5 baseline SD (non-response; label-using, diagnostic only). Together they buy 0.05–0.06 BA on PhysioNet and Stress-Predict (0.02 for the external model on Stress-Predict), and only F1 and F3 on Stress-Predict and F3 external on PhysioNet have confidence intervals excluding zero. The residual error of 0.17–0.23 (0.33 external on Stress-Predict) is explained by none of them. The responder filter is the one term sensitive to its definition (`partB_responder_sensitivity.csv`). At 0.5 or 1.0 baseline SD on either channel the two rules select the same subjects (32 of 35 PhysioNet, 32 of 34 Stress-Predict) and F4 gains 0.01–0.02. Requiring 1.0 SD on both channels keeps 17 of 35 PhysioNet subjects and raises within-dataset BA by 0.094 [0.05, 0.15] (external 0.065 [0.01, 0.13]); keeping the top half of subjects by pooled response gives 0.062 [0.01,

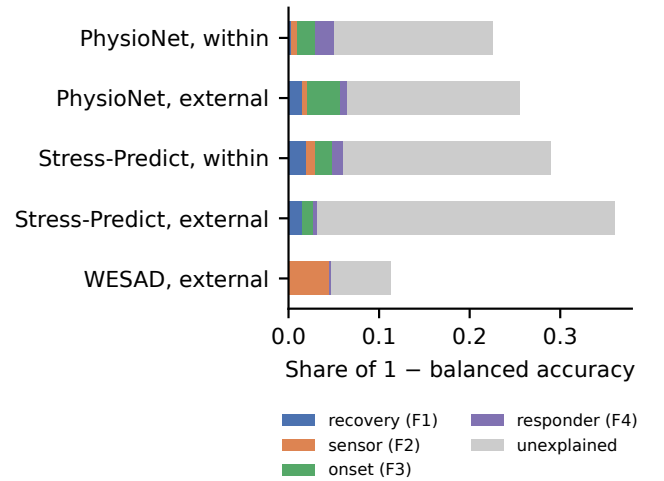


Fig. 4. Error budget. Each bar is 1–BA of the out-of-sample predictions; coloured segments are the Shapley-averaged gains of the four test-side filters, grey is the residual after all four. Negative gains (Stress-Predict external, F2) are drawn as zero.

0.13]. On Stress-Predict the strict rules keep 12–17 of 34 subjects and gain 0.03 within (CI includes zero) and −0.02 to +0.03 externally. Under the strict rules non-response is the largest single term on PhysioNet, about half of the four-filter total, yet the residual stays at 0.10–0.14 on PhysioNet and 0.21–0.35 on Stress-Predict. Onset latency and recovery are real but small: within-dataset Stress-Predict recall is 0.47 in the first minute of a stressor against 0.77 after 3 min, PhysioNet 0.63 in the first minute against 0.77 in the third, and rest false alarms peak in the first minute after offset (PhysioNet within 0.26 against 0.06–0.09 later); too few windows sit there to move BA.

TABLE VI
LEAKAGE-CONTROLLED NEGATIVE RESULTS (BALANCED ACCURACY,
HR/HRV/EDA UNLESS STATED)

Question	Result
Unsupervised domain adaptation (correlation alignment, maximum mean discrepancy, domain-adversarial and subject-adversarial networks), WESAD $\leftrightarrow$ PhysioNet, physiology	0.52–0.70 vs source-only 0.54–0.70; none reliably better
Few-shot on unseen dataset, first-in-time windows, $k = 1$ –5 per class	–0.04 to +0.04 refit gain; none significant after Holm
Few-shot, random windows	only significant gains (WESAD $k = 5$, +0.03); not deployable hurts PhysioNet (-0.08 , $p_{\text{holm}} < 0.05$ at $k = 3$)
Per-user threshold from calibration windows	PhysioNet $\rho = -0.08$ [–0.42, 0.21]
Self-rated stress rise vs window recall (PhysioNet 54 units, WESAD 15)	within, 0.02 [–0.30, 0.31] external; WESAD -0.49 [–0.87, 0.17] within, -0.60 [–0.84, -0.08] external (15 units); mild tasks not missed more
Detectability as a subject trait: recall across stressors	Stress-Predict $\rho = -0.14$ [–0.51, 0.24]; PhysioNet 0.44 [–0.04, 0.78]
Recall vs baseline HR, RMSSD, EDA, PPG coverage	$ \rho \leq 0.35$ with CIs including 0, except Stress-Predict external recall vs baseline EDA ($\rho = 0.48$ [0.18, 0.70])
Worst 20% of subjects' share of misses	37–61%
Exercise absent from training (LODO with exercise as negatives)	69.5% of exercise windows called stress by the other-4 model; 4–6% when PhysioNet's own subjects are in training (8–10% under the panel's added-negative configuration, Table III)

On WESAD the only sizeable term is the sensor (F2, +0.045 external). Replacing wrist cardiac features with chest-ECG features on the same windows raises BA from 0.916 to 0.956 (AUROC 0.977 to 0.993), and chest ECG alone reaches 0.918, equal to the whole wrist pipeline; the gain concentrates in low-coverage windows, where wrist heart rate is absent (0.886 to 0.962). EDA alone gives 0.891. Motion-robust wrist heart rate is therefore worth at most about +0.04 on WESAD, and about 0.04 of error remains with chest-grade cardiac data.

G. Negative results

Table VI collects results that close directions the literature has proposed. Unsupervised domain adaptation on top of per-subject normalisation does not beat source-only training in either direction between WESAD and PhysioNet, consistent with Kwon et al. [5]. Few-shot personalisation of a model trained on the other four datasets, with each user's first k windows per class as calibration and a 30-s gap before any test window, adds at most 0.02–0.04 BA and nothing significant; drawing calibration windows at random from the recording produces the only significant gains, in line with Tervonen et al. [18] and Akkaya [19]. A per-user threshold hurts. Self-rated stress intensity does not predict which windows are missed: in PhysioNet, Stroop (mean rise 0.7 of 10) is detected as well as arithmetic (2.5); WESAD's negative correlation rests on the two subjects with the largest rise. The data

give no support to detectability as a stable subject trait: recall is not correlated across stressors, is not predicted by baseline physiology or PPG coverage (one exception, Stress-Predict external recall against baseline EDA), and misses are only moderately concentrated; PhysioNet's within and external models do agree on who is hard ($\rho = 0.64$ [0.37, 0.82]).

VI. DISCUSSION

Wrist stress detectors behave as arousal detectors. Three results point the same way. An untrained index of heart rate and tonic EDA transfers across datasets as well as the trained model. At matched arousal, the model separates stress from baseline and rest only weakly (0.584), and no better from the lower-demand social task, hyperventilation, Lego assembly or anger clips. And the one clean non-stress state with high arousal, exercise, is called stress until it is shown as a negative. Once labels are protocol-anchored and features are normalised per subject, the distribution of that arousal is similar across laboratory datasets, so transfer is cheap and domain adaptation has little to align. This is a statement about what stage-labelled training extracts from wrist HR, HRV and EDA in these paradigms, not a claim that stress has no autonomic signature. Two readings remain open and the present data cannot separate them: the detector may have learned only arousal, or the wrist channels may carry nothing more: psychosocial stress activates the hypothalamic–pituitary–adrenal axis as well as the sympathetic system, and that axis is assessed through cortisol [9], which no wrist channel measures, while wrist EDA tracks palmar EDA only moderately [20]. Kwon et al. [5] found WESAD stress separable from baseline at matched heart rate through EDA; our reference row, which matches heart rate and EDA together, suggests that this residual is largely electrodermal arousal. Exercise and fear clips are the separable states, and the channel-disjoint variants show that exercise is separated by its cross-channel pattern (heart rate far above the electrodermal level) rather than by a stress signature. The datasets with low ceilings are those whose stressors produce a sixth (PhysioNet) to a third (Stress-Predict) of the electrodermal response of the TSST in WESAD; filters for recovery, sensor loss and onset recover at most 0.06 of that gap, and a strict responder filter at most 0.09 on PhysioNet and nothing on Stress-Predict. We read the ceiling as set by the size of the physiological response a protocol provokes. Because that size is measured with the same channels, this is a description, not an independent test of stressor potency. The few-shot and domain-adaptation nulls are consistent with it: personalisation cannot recover a response that did not occur.

Deployment. A detector that responds to arousal will raise alerts during any arousing activity. As an illustration, at a stress prevalence of 5% and recall 0.75, false-stress rates of 0.10, 0.33 and 0.69, the range of Table III, give positive predictive values of 0.28, 0.11 and 0.05. Outputs of such detectors are better labelled “arousal” than “stress” until a specificity panel says otherwise.

Reporting norms. We propose that a wearable stress detector be published with three checks: an index-only baseline, reporting how far the trained model beats an untrained HR+EDA index under the same split; a specificity panel, its false-stress rate

on exercise, on lower-demand versions of its stressor tasks and on strong emotion, with CIs and at least about 20 subjects per state, the way an assay reports cross-reactivity; and a matched-arousal test against a reference pair, with a stated equivalence margin, where such states are available. We also propose three protocol rules: a settling period before baseline, or time since donning as a covariate; position-matched negatives, or at least a report of the time-only classifier, whenever baseline precedes the stressor; and exercise among the training negatives, or an activity gate, whenever the deployment includes movement.

What a decisive study needs. No public E4 dataset separates speech from evaluation within subject or counterbalances order; UBFC-Phys contrasts two demand levels of a social task between subjects. The friendly TSST, built to match the TSST in structure and cognitive demand with the social-evaluative elements removed, and run within subject in randomised order by Richer et al. [9], is the control such a study needs on the wrist. A within-subject design with speech and evaluation crossed, cycling at matched heart rate, cognitive load without feedback, Latin-square order with baselines at both ends, chest ECG and a palmar EDA reference would settle whether wrist EDA carries an evaluative-threat signature that the present data cannot resolve, and would provide the first order-matched baseline on the wrist.

Limitations. Whole-session per-subject normalisation is transductive; raw and causal variants keep the small transfer cost on long recordings but lose about 0.10 on short ones. The probed states are not clean non-stress controls apart from exercise and the passive clips, the hyperventilation, anger and UBFC-Phys control cells are small, and the UBFC-Phys contrast is between subjects; the matched-arousal nulls exclude only deviations above 0.12–0.22. Mean skin-conductance level from the E4 correlated with palmar EDA at $r = 0.61$ and 0.40 in two resting baselines and not significantly ($r = 0.30$) during conversation [20], so a wrist null on a stress-specific signature may be instrument failure. The arousal index is built from two model features; the channel-disjoint variants keep hyperventilation and the UBFC-Phys control inseparable but make Lego or anger separable in one variant each, and heart rate is missing far more often in some stress classes (63% of WESAD stress windows) than in their baselines. Responder filters use test labels and are diagnostic only; the strict definitions discard half the subjects. All data are laboratory recordings; ambulatory validity is untested. The pooled subject count (132 in the five stress datasets) is small for a field-level claim, and the upper CI bounds of the transfer costs reach 0.08–0.13.

VII. CONCLUSION

Across five public laboratory E4 datasets with audited labels, a wrist stress detector built on HR, HRV and EDA features transfers to an unseen dataset at a cost of 0.015–0.05 BA, with no cost significant and upper bounds of 0.08–0.13, and none of the methods tested here, from domain adaptation to few-shot calibration, improved on that. An untrained index of heart rate and tonic EDA transfers as well. At matched arousal the detector separates stress from rest only weakly and

no better from lower-demand social tasks, hyperventilation, manual work or anger clips, and it calls exercise stress until exercise is in training. On these data, wrist stress detectors behave as arousal detectors. Protocol position predicts stress better than physiology on the hardest datasets, whose baselines are recorded while the sensor settles, and the within-dataset ceiling tracks the size of the physiological response each protocol provokes rather than the pipeline. The next gains are more likely to come from protocol design and contextual sensing than from classifiers. Two studies follow from this work: a counterbalanced within-subject protocol that crosses speech with evaluation, adds matched-heart-rate exercise and non-evaluative cognitive load, and records chest ECG and palmar EDA alongside the wrist, to test whether wrist EDA carries an evaluative-threat signature at all; and an ambulatory validation in which the specificity panel is scored against labelled daily-life arousal, which no laboratory dataset can provide.

DATA AND CODE AVAILABILITY

The five datasets are available from their original sources under their own licences. The audited label tables, the 20 frozen subject-grouped splits, the feature-extraction and evaluation scripts, and the scripts for the specificity panel, matched-arousal test, time-in-session probe and error budget are released at <https://github.com/ALVAROI12/smartwatch-stress-detection> (branch `jbhi-revision`).

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